MOMO Quant

Part 3 — Database Design Document

Version: 1.0 Draft
Database: MySQL
ORM: Entity Framework Core
Design Goal: Store enough data to backtest, replay, paper trade, report, audit, and debug every trading decision.

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1. Database Design Principle

The database must not only store trades.

It must store:

- market data
- indicators
- signals
- AI decisions
- risk decisions
- orders
- fills
- positions
- PnL
- rejected trades
- missed trades
- bot mode changes
- audit logs
- monitoring events

A trading bot that only stores winning/losing trades is useless for serious improvement.

MOMO Quant must store the full decision trail.

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2. Main Database Areas

The database will be divided into these logical areas:

```
1. Identity & Security
2. Exchange & Symbols
```

```
3. Market Data
4. Indicators
5. Strategies
6. AI Decisions
7. Risk Management
8. Trading Sessions
9. Signals
10. Orders
11. Trades
12. Positions
13. Backtesting
14. Replay
15. Paper Trading
16. Reporting
17. Monitoring
18. Audit Logs
19. Settings
```

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3. Naming Rules

Use:

- PascalCase for C# entities.
- snake_case or PascalCase consistently in MySQL.
- UTC timestamps only.
- Decimal for all financial values.
- Never use floating point for price, quantity, or PnL.

Recommended C# decimal precision:

```
decimal(28, 12)
```

For money and price fields.

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4. Identity & Security Tables

4.1 Users

Stores system users.

Fields:

```
Id
FullName
Email
PasswordHash
RoleId
IsActive
LastLoginAt
CreatedAt
UpdatedAt
```

4.2 Roles

Fields:

```
Id
Name
Description
CreatedAt
UpdatedAt
```

Initial roles:

```
Admin
Trader
Viewer
```

4.3 ApiKeyVault

Stores encrypted exchange API keys.

Fields:

```
Id
ExchangeId
Name
ApiKeyEncrypted
ApiSecretEncrypted
PassphraseEncrypted
IsTestnet
```

```
IsActive
CreatedAt
UpdatedAt
```

Never store raw API keys.

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5. Exchange & Symbol Tables

5.1 Exchanges

Fields:

```
Id
Name
Code
BaseUrl
WebSocketUrl
IsActive
CreatedAt
UpdatedAt
```

Examples:

```
Binance Futures
Bybit
OKX
```

MVP should support one exchange first.

5.2 Symbols

Fields:

```
Id
ExchangeId
Symbol
BaseAsset
QuoteAsset
ContractType
PricePrecision
```

```
QuantityPrecision
MinQty
MinNotional
TickSize
StepSize
MakerFeeRate
TakerFeeRate
IsActive
CreatedAt
UpdatedAt
```

Example symbols:

```
BTCUSDT
ETHUSDT
SOLUSDT
BNBUSDT
XRPUSDT
```

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6. Market Data Tables

6.1 Candles

Stores OHLCV data.

Fields:

```
Id
ExchangeId
SymbolId
Timeframe
OpenTimeUtc
CloseTimeUtc
Open
High
Low
Close
Volume
QuoteVolume
TradeCount
```

```
IsClosed
CreatedAt
```

Indexes:

```
ExchangeId + SymbolId + Timeframe + OpenTimeUtc unique
SymbolId + Timeframe + OpenTimeUtc
OpenTimeUtc
```

This is one of the most important tables.

6.2 OrderBookSnapshots

Optional for later.

Fields:

```
Id
ExchangeId
SymbolId
CapturedAtUtc
BestBidPrice
BestBidQty
BestAskPrice
BestAskQty
Spread
MidPrice
CreatedAt
```

MVP can delay this table.

6.3 FundingRates

Fields:

```
Id
ExchangeId
SymbolId
FundingTimeUtc
FundingRate
```

```
MarkPrice
CreatedAt
```

Useful for futures reporting.

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7. Indicator Tables

7.1 IndicatorSnapshots

Stores calculated indicator values per candle.

Fields:

```
Id
SymbolId
Timeframe
CandleId
CalculatedAtUtc
Ema20
Ema50
Ema200
Vwap
Rsi14
Atr14
VolumeSma20
SwingHigh
SwingLow
MarketStructure
CreatedAt
```

Indexes:

```
SymbolId + Timeframe + CandleId unique
SymbolId + Timeframe + CalculatedAtUtc
```

Do not calculate indicators only in memory if replay/reporting needs them later.

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8. Strategy Tables

8.1 Strategies

Fields:

```
Id
Code
Name
Description
IsEnabled
Version
CreatedAt
UpdatedAt
```

Initial strategy codes:

```
LIQUIDITY_SWEEP
VWAP_MEAN_REVERSION
EMA_PULLBACK
```

8.2 StrategyParameters

Stores configurable strategy parameters.

Fields:

```
Id
StrategyId
ParameterKey
ParameterValue
ValueType
Timeframe
SymbolId nullable
IsActive
CreatedAt
UpdatedAt
```

Examples:

```
EmaFastPeriod = 20
EmaSlowPeriod = 50
AtrMultiplier = 1.5
MinConfidence = 80
```

Symbol-specific overrides are allowed.

---

9. AI Decision Tables

9.1 AiDecisions

Stores AI outputs for every evaluated signal or market state.

Fields:

```
Id
TradingSessionId
SymbolId
Timeframe
CandleId nullable
SignalId nullable
MarketRegime
ConfidenceScore
PreferredStrategyCode
RiskAdjustment
TradeAllowed
Explanation
RawRequestJson
RawResponseJson
CreatedAt
```

Indexes:

```
TradingSessionId
SymbolId + Timeframe + CreatedAt
SignalId
```

This table is critical for explainability.

9.2 AiModelVersions

Fields:

```
Id
ModelName
ModelVersion
Purpose
TrainingDataFrom
TrainingDataTo
MetricsJson
IsActive
CreatedAt
```

Use this later when models become more advanced.

---

10. Risk Management Tables

10.1 RiskProfiles

Fields:

```
Id
Name
Description
IsDefault
CreatedAt
UpdatedAt
```

Examples:

```
Conservative
Balanced
Aggressive
```

10.2 RiskRules

Fields:

```
Id
RiskProfileId
RuleKey
RuleValue
ValueType
IsEnabled
CreatedAt
UpdatedAt
```

Examples:

```
MaxRiskPerTradePercent = 0.5
MaxDailyLossPercent = 2
MaxOpenPositions = 2
MaxConsecutiveLosses = 3
MinConfidenceScore = 80
MaxSpreadPercent = 0.05
```

10.3 RiskDecisions

Stores risk engine decisions.

Fields:

```
Id
TradingSessionId
SignalId nullable
AiDecisionId nullable
SymbolId
Decision
Reason
ApprovedRiskPercent
PositionSize
StopLoss
TakeProfit
RejectedRuleKey
CreatedAt
```

Decision values:

```
Approved
Rejected
Adjusted
EmergencyBlocked
```

---

11. Trading Session Tables

11.1 TradingSessions

Every backtest, replay, paper run, or live run gets a session.

Fields:

```
Id
Name
Mode
Status
ExchangeId
StartedByUserId
StartedAtUtc
StoppedAtUtc
InitialBalance
FinalBalance
Notes
CreatedAt
UpdatedAt
```

Mode values:

```
Backtest
Replay
Paper
Live
```

Status values:

```
Created
Running
Paused
Stopped
```

```
Completed
Failed
```

This table ties everything together.

11.2 TradingSessionSymbols

Fields:

```
Id
TradingSessionId
SymbolId
Timeframe
HigherTimeframe
IsActive
CreatedAt
```

---

12. Signal Tables

12.1 StrategySignals

Stores all strategy signals, including rejected and ignored ones.

Fields:

```
Id
TradingSessionId
StrategyId
SymbolId
Timeframe
CandleId nullable
SignalType
Direction
Strength
ConfidenceContribution
EntryPrice
SuggestedStopLoss
SuggestedTakeProfit
Reason
```

```
RawDataJson
CreatedAt
```

SignalType:

```
Entry
Exit
Warning
NoTrade
```

Direction:

```
Long
Short
None
```

Important: store NoTrade signals when useful for debugging.

---

13. Order Tables

13.1 Orders

Stores order intent and lifecycle.

Fields:

```
Id
TradingSessionId
SymbolId
TradeId nullable
ExternalOrderId nullable
Mode
Side
OrderType
PositionSide
Price
Quantity
Status
IsPostOnly
IsReduceOnly
TimeInForce
```

```
RequestedAtUtc
SubmittedAtUtc
CancelledAtUtc
FilledAtUtc
FailureReason
CreatedAt
UpdatedAt
```

Side:

```
Buy
Sell
```

OrderType:

```
Limit
Market
StopLimit
StopMarket
```

Status:

```
Created
Submitted
PartiallyFilled
Filled
Cancelled
Rejected
Expired
Failed
```

13.2 OrderFills

Stores partial/full fills.

Fields:

```
Id
OrderId
ExternalFillId nullable
FillPrice
```

```
FillQuantity
Fee
FeeAsset
LiquidityType
FilledAtUtc
CreatedAt
```

LiquidityType:

```
Maker
Taker
SimulatedMaker
SimulatedTaker
```

This is required for realistic performance analysis.

13.3 MissedOrders

Stores valid signals that did not fill due to maker limit logic.

Fields:

```
Id
TradingSessionId
SignalId
SymbolId
RequestedPrice
BestBid
BestAsk
Reason
ExpiredAtUtc
CreatedAt
```

Examples:

```
Maker order not filled
Price moved away
Spread too wide
Order timeout reached
```

---

14. Trade Tables

14.1 Trades

Stores completed or active trade lifecycle.

Fields:

```
Id
TradingSessionId
SymbolId
StrategyId nullable
SignalId nullable
AiDecisionId nullable
RiskDecisionId nullable
Direction
EntryOrderId nullable
ExitOrderId nullable
EntryPrice
ExitPrice nullable
Quantity
StopLoss
TakeProfit
Status
OpenedAtUtc
ClosedAtUtc nullable
GrossPnl
Fees
FundingFees
NetPnl
RMultiple
CloseReason
CreatedAt
UpdatedAt
```

Status:

```
Open
Closed
Cancelled
Failed
```

CloseReason:

```
TakeProfit
StopLoss
ManualClose
StrategyExit
RiskExit
EmergencyStop
Timeout
```

---

15. Position Tables

15.1 Positions

Tracks current position state.

Fields:

```
Id
TradingSessionId
SymbolId
Direction
Quantity
AverageEntryPrice
MarkPrice
UnrealizedPnl
RealizedPnl
Leverage
MarginUsed
Status
OpenedAtUtc
UpdatedAtUtc
ClosedAtUtc nullable
```

Status:

```
Open
Closed
Liquidated
Error
```

---

16. Backtesting Tables

16.1 BacktestRuns

Fields:

```
Id
TradingSessionId
Name
SymbolId
Timeframe
HigherTimeframe
StartDateUtc
EndDateUtc
InitialBalance
FinalBalance
StrategySetJson
SettingsJson
Status
StartedAtUtc
CompletedAtUtc
CreatedAt
```

16.2 BacktestResults

Fields:

```
Id
BacktestRunId
TotalTrades
WinningTrades
LosingTrades
WinRate
GrossPnl
NetPnl
TotalFees
MaxDrawdown
ProfitFactor
Expectancy
AverageWin
AverageLoss
LargestWin
```

```
LargestLoss
SharpeRatio
SortinoRatio
CreatedAt
```

---

17. Replay Tables

17.1 ReplaySessions

Fields:

```
Id
TradingSessionId
SymbolId
Timeframe
StartTimeUtc
EndTimeUtc
CurrentReplayTimeUtc
ReplaySpeed
Status
CreatedAt
UpdatedAt
```

Replay state can also be stored in Redis for performance.

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18. Paper Trading Tables

18.1 PaperAccounts

Fields:

```
Id
Name
InitialBalance
CurrentBalance
Currency
IsActive
CreatedAt
UpdatedAt
```

18.2 PaperAccountSnapshots

Fields:

```
Id
PaperAccountId
TradingSessionId
Balance
Equity
UnrealizedPnl
RealizedPnl
MarginUsed
CapturedAtUtc
CreatedAt
```

19. Reporting Tables

Most reports can be generated from trades/orders/signals.

Optional materialized report tables:

19.1 DailyPerformanceSnapshots

Fields:

```
Id
TradingSessionId
DateUtc
StartingBalance
EndingBalance
GrossPnl
NetPnl
Fees
TradeCount
WinRate
MaxDrawdown
CreatedAt
```

19.2 StrategyPerformanceSnapshots

Fields:

```
Id
TradingSessionId
StrategyId
SymbolId nullable
TotalTrades
WinRate
NetPnl
ProfitFactor
MaxDrawdown
CreatedAt
```

---

20. Monitoring Tables

20.1 SystemHealthLogs

Fields:

```
Id
ServiceName
Status
Message
LatencyMs
CheckedAtUtc
CreatedAt
```

Status:

```
Healthy
Warning
Critical
Stopped
```

20.2 ExchangeConnectionLogs

Fields:

```
Id
ExchangeId
ConnectionType
Status
Message
ReconnectCount
LatencyMs
CheckedAtUtc
CreatedAt
```

ConnectionType:

```
Rest
WebSocket
```

---

21. Audit Logs

21.1 AuditLogs

Fields:

```
Id
UserId nullable
TradingSessionId nullable
Action
EntityType
EntityId nullable
OldValueJson
NewValueJson
IpAddress
UserAgent
CreatedAt
```

Audit these actions:

- mode changed
- live mode enabled
- risk rule changed
- strategy parameter changed
- API key changed
- bot started

- bot stopped
- emergency stop triggered

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22. Settings Tables

22.1 AppSettings

Fields:

```
Id
SettingKey
SettingValue
ValueType
Category
IsSensitive
CreatedAt
UpdatedAt
```

Examples:

```
DefaultTradingMode
EnabledSymbols
DefaultTimeframe
AiServiceUrl
MaxReplaySpeed
```

Sensitive settings should be encrypted or kept outside DB.

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23. Suggested Entity Relationships

Important relationships:

```
Exchange 1 -> many Symbols
TradingSession 1 -> many StrategySignals
TradingSession 1 -> many Orders
TradingSession 1 -> many Trades
TradingSession 1 -> many AiDecisions
TradingSession 1 -> many RiskDecisions
Strategy 1 -> many StrategySignals
Strategy 1 -> many Trades
```

```
Signal 1 -> many AiDecisions
Signal 1 -> many RiskDecisions
Trade 1 -> many Orders
Order 1 -> many OrderFills
BacktestRun 1 -> one BacktestResult
```

---

24. Database Implementation Order

Do not create every table immediately.

Build in this order:

```
1. Users / Roles
2. Exchanges / Symbols
3. Candles
4. Strategies / StrategyParameters
5. TradingSessions
6. IndicatorSnapshots
7. StrategySignals
8. AiDecisions
9. RiskDecisions
10. Orders / OrderFills / MissedOrders
11. Trades / Positions
12. BacktestRuns / BacktestResults
13. ReplaySessions
14. PaperAccounts / PaperAccountSnapshots
15. Reports
16. Monitoring
17. AuditLogs
18. Settings
```

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25. Critical Database Rules

1. Use UTC for every timestamp.
2. Use decimal, never float, for trading values.
3. Store every decision, not only executed trades.
4. Keep raw AI request/response JSON for debugging.
5. Store missed orders separately.
6. Store rejected risk decisions.
7. Never store plain exchange API keys.
8. Design reports from stored facts, not assumptions.

9. Do not delete trading records; mark them inactive or cancelled.
10. Every live-mode action must be auditable.

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26. MVP Database Scope

For MVP, create these first:

```
Users
Roles
Exchanges
Symbols
Candles
Strategies
StrategyParameters
TradingSessions
IndicatorSnapshots
StrategySignals
AiDecisions
RiskRules
RiskDecisions
Orders
OrderFills
MissedOrders
Trades
Positions
BacktestRuns
BacktestResults
ReplaySessions
PaperAccounts
PaperAccountSnapshots
AuditLogs
SystemHealthLogs
AppSettings
```

Delay these:

```
OrderBookSnapshots
FundingRates
AiModelVersions
StrategyPerformanceSnapshots
DailyPerformanceSnapshots
ExchangeConnectionLogs
```

27. Final Database Summary

The MOMO Quant database must support research, execution, reporting, and accountability.

The most important design decision is this:

The system must store not only what it traded, but also what it considered trading and why it rejected or missed trades.

Without that, the bot cannot be improved intelligently.